

Project Title:

Lightweight Confidence-Aware Semi-Supervised Network Intrusion Detection System

Using Contrastive Feature Learning

Degree/Program: B.Tech Computer Science Engineering

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Abstract:

Network Intrusion Detection Systems (NIDS) play a crucial role in protecting modern networks from malicious activities; however, existing approaches often depend on large volumes of labeled data and exhibit poor performance in detecting minority and novel attacks. Semi-supervised learning methods partially address this issue by leveraging unlabeled data, but clustering-based pseudo-labeling techniques frequently suffer from error propagation and limited generalization.

This project proposes a lightweight confidence-aware semi-supervised NIDS that combines contrastive feature learning with reliable pseudo-label selection to improve intrusion detection performance. The proposed framework learns robust and discriminative feature representations from network traffic using contrastive learning and selectively incorporates unlabeled samples based on confidence thresholds. A feature refinement stage and ensemble-based classifier further enhance detection accuracy and robustness.

Experimental evaluation on benchmark datasets demonstrates that the proposed approach improves overall detection performance, enhances minority attack detection, and reduces false positives compared to existing semi-supervised intrusion detection methods. The lightweight design also makes the system suitable for practical deployment in resource-constrained network environments.

Introduction:

The rapid growth of computer networks, cloud infrastructure, and Internet of Things (IoT) environments has significantly increased the complexity and volume of network traffic. Alongside this growth, cyber-attacks such as denial-of-service attacks, probing, malware exploitation, and unauthorized access attempts have become more frequent and sophisticated. Network Intrusion Detection Systems (NIDS) play a critical role in monitoring network traffic and identifying malicious activities to ensure the security and reliability of modern digital infrastructures.

Traditional intrusion detection approaches, including signature-based and rule-based systems, are effective for detecting known attack patterns but struggle to identify previously unseen or evolving threats. These systems require frequent manual updates and lack the adaptability needed for dynamic network environments. As a result, machine learning-based NIDS have gained attention due to their ability to automatically learn patterns from data and detect complex attack behaviors.

Despite their advantages, machine learning and deep learning-based intrusion detection systems face several challenges. One major limitation is the heavy reliance on large volumes of labeled network traffic data, which is expensive and time-consuming to obtain. In real-world scenarios, most network traffic remains unlabeled, making fully supervised learning approaches impractical. Additionally, network intrusion datasets are highly imbalanced, with normal traffic and certain attack types dominating the data, while minority and novel attacks remain underrepresented. This imbalance often leads to poor detection performance for critical but rare attack classes.

To address the scarcity of labeled data, semi-supervised learning techniques have been introduced to leverage both labeled and unlabeled network traffic. However, many existing semi-supervised NIDS rely on clustering-based pseudo-labeling methods, which assume well-separated data distributions. In practice, real network traffic is noisy, heterogeneous, and continuously evolving, causing these methods to generate unreliable labels and propagate errors during training. Furthermore, conventional feature extraction techniques, such as handcrafted features or basic autoencoders, often fail to capture complex and discriminative representations necessary for detecting subtle and evolving attack patterns.

Motivated by these limitations, this project proposes a lightweight confidence-aware semi-supervised Network Intrusion Detection System that integrates contrastive feature learning with reliable pseudo-label selection. The proposed approach aims to learn robust and discriminative feature representations from network traffic while selectively incorporating unlabeled samples based on confidence thresholds. By reducing error propagation and enhancing minority and novel attack detection, the system seeks to improve overall intrusion detection performance while maintaining computational efficiency. This makes the proposed framework suitable for practical deployment in real-world and resource-constrained network environments.

Problem Statement:

Modern network environments generate large volumes of high-dimensional and continuously evolving traffic, making intrusion detection a challenging task. Existing Network Intrusion Detection Systems (NIDS) largely rely on fully supervised learning techniques that require extensive labeled datasets, which are costly and impractical to obtain in real-world scenarios. Although semi-supervised approaches attempt to utilize unlabeled data, many existing methods depend on clustering-based pseudo-labeling, leading to unreliable label assignments and error propagation, particularly under highly imbalanced data distributions.

Furthermore, current intrusion detection models struggle to accurately identify minority and novel attack patterns, which are often the most critical from a security perspective. Static feature representations and rigid decision boundaries further limit adaptability to evolving attack behaviors. Therefore, there is a need for a robust, efficient, and semi-supervised intrusion detection framework that can reliably leverage unlabeled network traffic, learn discriminative feature representations, and improve detection performance for minority and previously unseen attacks while maintaining computational efficiency.

Research Objectives:

Primary Objective

To design and evaluate a lightweight confidence-aware semi-supervised Network Intrusion Detection System that effectively utilizes unlabeled network traffic and improves the detection of minority and novel attacks.

Specific Objectives

1. To analyze the limitations of existing supervised and semi-supervised intrusion detection systems with respect to labeled data dependency and class imbalance.
2. To develop a semi-supervised learning strategy that incorporates unlabeled network traffic using confidence-based pseudo-label selection to reduce error propagation.
3. To learn robust and discriminative feature representations from network traffic using contrastive feature learning techniques.
4. To enhance intrusion detection performance for minority and previously unseen attack patterns under imbalanced data conditions.
5. To evaluate the performance of the proposed system using standard metrics such as accuracy, precision, recall, and F1-score.
6. To compare the proposed approach with baseline intrusion detection methods to assess improvements in detection effectiveness and robustness.

Hypotheses:

Hypothesis 1

A confidence-aware semi-supervised learning approach will improve intrusion detection performance compared to traditional semi-supervised methods that use clustering-based pseudo-labeling.

Hypothesis 2

Contrastive feature learning will produce more discriminative feature representations than conventional autoencoder-based feature extraction for network intrusion detection.

Hypothesis 3

The proposed framework will achieve higher detection performance for minority and novel attack classes under imbalanced data conditions compared to baseline intrusion detection models.

Hypothesis 4

The lightweight design of the proposed system will maintain competitive detection performance while reducing computational complexity, making it suitable for practical deployment.

Literature Review:

Network Intrusion Detection Systems (NIDS) have been widely studied as a critical component of network security frameworks. With the rapid growth of networked systems, traditional rule-based and signature-based intrusion detection techniques have become increasingly ineffective in handling large-scale, dynamic, and complex network traffic. This limitation has led to the adoption of machine learning and deep learning techniques for automated intrusion detection.

2.1 Traditional Machine Learning-Based Intrusion Detection

Early machine learning-based NIDS employed supervised learning algorithms such as Decision Trees, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests to classify network traffic as either normal or malicious. Studies by Belavagi and Muniyal demonstrated that ensemble methods like Random Forest outperform individual classifiers in terms of accuracy and robustness. However, these approaches rely heavily on handcrafted features and fully labeled datasets, which limits their effectiveness in real-world environments where labeled intrusion data is scarce and attack patterns evolve over time.

Feature selection techniques have been explored to improve the performance of traditional machine learning models. Kasongo and Sun showed that optimized feature subsets can enhance detection accuracy; however, their approach still struggled with detecting minority attack classes under imbalanced data distributions. These findings highlight that while traditional machine learning approaches are computationally efficient, their dependency on labeled data and static feature representations restricts their generalization capabilities.

2.2 Deep Learning-Based Intrusion Detection Systems

To overcome the limitations of shallow learning models, deep learning techniques such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and hybrid architectures have been proposed for intrusion detection. CNN-based models are effective in capturing spatial patterns in network traffic, while LSTM-based models are capable of learning temporal dependencies. Du et al. introduced a CNN-LSTM-based NIDS that achieved improved detection performance on benchmark datasets. Similarly, CNN-BiLSTM architectures have been shown to enhance detection accuracy for known attack types.

Autoencoder-based models have also been widely adopted for dimensionality reduction and anomaly detection. By learning compact latent representations of network traffic, autoencoders help reduce noise and redundancy in high-dimensional data. However, most autoencoder-based approaches focus on reconstruction error and static feature representations, which limits their ability to detect complex, evolving, and time-dependent attack behaviors. Additionally, deep learning-based NIDS often require large labeled datasets and suffer from class imbalance, leading to poor detection of rare and novel attacks.

2.3 Semi-Supervised Learning for Intrusion Detection

Given the high cost of labeling network traffic data, semi-supervised learning has emerged as a promising direction for NIDS. Semi-supervised approaches aim to leverage a small amount of labeled data together with a large volume of unlabeled data. Clustering-based methods, such as those using DBSCAN, have been employed to generate pseudo-labels for unlabeled samples. Li et al. proposed a semi-supervised NIDS that combines autoencoders, DBSCAN clustering, and ensemble classifiers, achieving improved performance compared to fully supervised methods.

Despite their effectiveness, clustering-based semi-supervised methods assume well-defined data density and separability in the feature space. In real-world network environments, traffic data is noisy, heterogeneous, and highly imbalanced, causing clustering algorithms to misclassify minority and novel attacks as noise. Incorrect pseudo-label assignments can propagate errors during training, leading to degraded detection performance. This highlights the need for more reliable mechanisms to incorporate unlabeled data into intrusion detection models.

2.4 Feature Representation Learning and Research Gaps

Feature representation learning plays a critical role in the success of intrusion detection systems. While handcrafted features require domain expertise and lack adaptability, deep learning-based feature extraction methods provide better representation capabilities. Recent studies have explored self-supervised and contrastive learning techniques to learn discriminative representations without relying on labels. Contrastive learning has shown strong performance in learning robust embeddings in various domains; however, its application in semi-supervised network intrusion detection remains limited.

Furthermore, most existing NIDS frameworks are static and lack mechanisms to adapt to evolving attack patterns and concept drift. Ensemble-based classifiers improve robustness but still rely on fixed decision boundaries. These limitations indicate a research gap in developing a lightweight, confidence-aware semi-supervised intrusion detection framework that combines robust feature representation learning with reliable pseudo-label selection to improve minority and novel attack detection.

Proposed Methodology:

This research proposes a lightweight confidence-aware semi-supervised Network Intrusion Detection System (NIDS) designed to effectively utilize both labeled and unlabeled network traffic while improving the detection of minority and novel attack patterns. The methodology follows a modular pipeline architecture, where each stage contributes to improved feature representation, reliable learning from unlabeled data, and robust intrusion classification.

3.1 Data Collection and Preprocessing

The proposed system is evaluated using standard benchmark intrusion detection datasets, including NSL-KDD and UNSW-NB15. Raw network traffic features are first preprocessed to ensure consistency and quality. Preprocessing steps include removal of irrelevant attributes, encoding of categorical features, normalization of numerical attributes, and label encoding for attack categories.

Given the highly imbalanced nature of intrusion detection datasets, class imbalance handling techniques such as controlled oversampling of minority classes and undersampling of majority classes are applied to reduce bias during training.

3.2 Feature Representation Learning using Contrastive Learning

To overcome the limitations of handcrafted and reconstruction-based feature extraction, the proposed framework employs contrastive feature learning to obtain robust and discriminative representations of network traffic.

A lightweight encoder network is trained using a contrastive learning objective, which encourages samples with similar behavior to have closer representations while pushing dissimilar samples apart in the latent space. This approach enables the model to learn meaningful features without requiring extensive labeled data and improves the separability between normal traffic and attack patterns.

The learned latent representations serve as the foundation for subsequent semi-supervised learning and classification stages.

3.3 Confidence-Aware Semi-Supervised Learning

In order to leverage unlabeled network traffic effectively, a confidence-aware pseudo-labeling strategy is incorporated into the training process. Instead of assigning pseudo-labels to all unlabeled samples, the model selectively assigns labels only to those samples for which the prediction confidence exceeds a predefined threshold.

This selective pseudo-labeling mechanism reduces the risk of propagating incorrect labels and improves the reliability of the semi-supervised learning process. Labeled and confidently pseudo-labeled samples are then combined to refine the decision boundaries and enhance generalization.

3.4 Feature Refinement and Intrusion Classification

Following semi-supervised learning, a feature refinement stage is applied to further enhance discriminative capability. A lightweight neural network is used to capture non-linear relationships among latent features and improve class separability.

The refined features are subsequently passed to an ensemble-based classifier, such as a Random Forest, which performs final intrusion detection and attack classification. Ensemble learning is adopted due to its robustness, resistance to overfitting, and ability to handle high-dimensional feature spaces.

3.5 Model Evaluation and Performance Analysis

The performance of the proposed NIDS is evaluated using standard metrics including accuracy, precision, recall, and F1-score. Class-wise evaluation is conducted to assess detection performance for minority and novel attack categories.

Comparative experiments are performed against baseline supervised and semi-supervised intrusion detection models to demonstrate the effectiveness of the proposed approach. Ablation studies may also be conducted to analyze the contribution of individual components within the framework.

Expected Results:

The proposed confidence-aware semi-supervised Network Intrusion Detection System is expected to improve intrusion detection performance compared to existing supervised and clustering-based semi-supervised approaches. By selectively incorporating unlabeled network traffic through confidence-based pseudo-labeling, the system is anticipated to reduce the impact of noisy labels and improve learning stability.

The use of contrastive feature learning is expected to generate more discriminative feature representations, enabling better separation between normal traffic and attack patterns, particularly for minority and previously unseen attacks. As a result, the proposed framework is expected to demonstrate improved precision, recall, and F1-score while maintaining competitive overall accuracy.

In addition, the lightweight design of the proposed approach is expected to ensure computational efficiency and practical feasibility. Overall, the expected results aim to validate the effectiveness of the proposed methodology in addressing key limitations of existing intrusion detection systems without introducing excessive computational complexity.

Significance of the Study:

This study is significant as it addresses key challenges in modern Network Intrusion Detection Systems related to labeled data scarcity, class imbalance, and poor detection of minority and novel attack patterns. By proposing a confidence-aware semi-supervised learning framework, the study contributes toward more reliable utilization of unlabeled network traffic, which reflects real-world network conditions where labeled data is limited.

From a technical perspective, the integration of contrastive feature learning enhances the quality of feature representations, enabling improved discrimination between normal and malicious traffic. This contributes to ongoing research efforts aimed at improving feature learning and generalization in intrusion detection systems without increasing reliance on extensive labeled datasets.

Practically, the lightweight design of the proposed framework makes it suitable for deployment in real-world and resource-constrained environments such as enterprise networks and IoT systems. Improved detection of minority and evolving attacks can enhance network security by reducing false negatives and strengthening early threat detection.

Academically, this study provides a structured approach for evaluating semi-supervised intrusion detection methods and serves as a foundation for future research in adaptive and scalable network security solutions. The findings may also support further exploration into confidence-aware and self-supervised learning techniques for cybersecurity applications.

Project Timeline:

Week	Phase	Detailed Activities
Week 1	Planning and Literature Refinement	Refine literature review and identify research gaps • Finalize problem statement, objectives, hypotheses, and methodology • Select benchmark datasets (NSL-KDD, UNSW-NB15) • Set up development environment (Python, ML libraries) • Prepare data understanding and initial exploratory analysis
Week 2	Data Preprocessing and Baseline Models	Data cleaning and removal of irrelevant features • Encoding of categorical attributes and feature normalization • Handling class imbalance using oversampling and undersampling techniques • Implement baseline supervised and semi-supervised intrusion detection models • Establish baseline performance metrics for comparison
Week 3	Contrastive Feature Learning Module	Design and implement lightweight contrastive encoder network • Train the model to learn discriminative latent representations from network traffic • Validate quality of learned representations using clustering and visualization techniques

		• Store learned feature embeddings for downstream learning
Week 4	Confidence Aware Semi-Supervised Learning	Implement confidence-based pseudo-label selection strategy • Integrate labeled and high-confidence unlabeled samples into training • Develop feature refinement network to improve class separability • Combine refined features with ensemble-based intrusion classifier
Week 5	Evaluation, Analysis and Documentation	Evaluate system performance using accuracy, precision, recall, and F1-score • Perform class-wise analysis focusing on minority and novel attacks • Compare proposed system with baseline models • Conduct ablation analysis to assess contribution of key components • Document findings, prepare final report and presentation

Tools and Resources:

The proposed system will be implemented using standard and widely adopted machine learning tools to ensure feasibility, reproducibility, and research validity.

Programming Language:

Python will be used as the primary programming language due to its extensive support for machine learning and data analysis.

Libraries and Frameworks:

- **PyTorch** for implementing contrastive feature learning and neural network components
- **Scikit-learn** for ensemble classifiers, baseline models, and performance evaluation
- **NumPy** and **Pandas** for numerical computation and dataset preprocessing
- **Imbalanced-learn (imblearn)** for handling class imbalance using oversampling and undersampling techniques
- **Matplotlib / Seaborn** for visualization of performance metrics and analysis results

Datasets:

Benchmark datasets such as **NSL-KDD** and **UNSW-NB15** will be used for training and evaluation.

Development Environment:

The system will be developed in a standard CPU-based environment using Python IDEs or Jupyter Notebook. Version control will be maintained using Git to ensure reproducibility.

Ethical Considerations:

This study is conducted using publicly available benchmark datasets for network intrusion detection, such as NSL-KDD and UNSW-NB15. These datasets contain preprocessed network traffic records and do not include personally identifiable information, ensuring that individual privacy is not compromised.

The proposed system is designed strictly for defensive cybersecurity purposes, aiming to improve the detection of malicious network activities and enhance network security. No attempt is made to misuse the methodology for offensive or intrusive actions. All experiments are performed in an offline and controlled environment, without interaction with real-time or live network traffic.

Additionally, care is taken to ensure responsible reporting of results by avoiding exaggerated claims and acknowledging limitations of the proposed approach. The study adheres to principles of academic integrity by properly citing existing literature and ensuring reproducibility of experiments.

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